Modeling the HRF with simultaneous EEG and fNIRS: conclusions

gradCPT_NN24 — status as of 2026-11-03

Section references (§) point to `summary_EEG_modeling_overview.ipynb` in `code/eeg/modeling/`, which holds the figures, tables and methods behind each statement.

Goal. The aim was to use simultaneous scalp EEG to explain or inform the fNIRS HbO response in gradCPT, in two ways:

- **trial-wise**, with single-trial Cz EEG features (ERP area, alpha power) modulating an event-locked HRF;
- **continuously**, with the Cz EEG as a delayed regressor, which gives an EEG→HbO transfer function.

Both were tried in channel and parcel space, with several noise models (AR-IRLS, fixAR-IRLS), a low-rank multi-parcel estimator (AR-iRRR) and a Bayesian joint HRF + AR model.

1. Findings

1. There is an event-locked HbO response to correct mountain withholds, and it does not need EEG.

- In ContA it peaks at ~4.5–6.5 s and dips at ~8 s (§4 Stage 7). Prefrontal control, default and ventral-attention parcels are the most significant, several of them in every subject (§4 Stage 8). The single-parcel example in §5g shows the same shape.
- The shape is robust to the noise model (AR-IRLS vs fixAR-IRLS r = 0.87–0.91) and to pipeline choices (r = 0.91–0.98).
- It is still weak relative to the noise at the single-subject level. Trained on one run, it does not improve prediction of the other two (best held-out iRRR rank 0 in 14 of 20 subjects).

2. EEG adds no detectable, generalizable information about HbO.

- **Trial-wise:** scaling the event regressors by the single-trial Cz ERP area leaves the HRF unchanged (r = 0.99, amplitude ratio 1.01). The ERP-area term is at chance (median 5% of parcels with $p < 0.05$). Its regressor correlates 0.70–0.92 with the plain event regressor, so the design has little leverage to detect a modulation (§5).
- **Continuous:**
  - The Cz-EEG regressors explain < 2% of the whitened HbO variance in-sample and ≈ 0 out-of-sample. Cross-validated λ_1 selects rank 0 for half the subjects (§4 Stages 5–6).
  - The resulting "HRFs" differ from the event-locked HRF (per-parcel r = 0.03–0.12, network-level shapes mostly anti-correlated). They often peak within 0–2 s of the EEG, which is too early for a hemodynamic response, and they vary across subjects.
  - Restricting the EEG to mnt-correct trials or using the whole run gives only partly similar estimates (r ≈ 0.45) (§5).
- **Significance without generalization:** large fractions of parcels pass F-tests in the continuous-EEG models (68–83%), yet those models fail on held-out runs. Parametric F-tests

with a slowly varying regressor and a shared whitening kernel look anti-conservative here.

3. Estimator choices change amplitude, not conclusions.

- fixAR-IRLS HRFs are ~0.8× the AR-IRLS amplitude, with nearly the same shape.
- AR-iRRR shrinks them to ~0.2–0.6× and smooths them.
- The low-rank components differ between subjects, so there is **no single shared EEG→HbO HRF shape** across subjects.

4. The Bayesian joint model points the same way. At DorsAttnB_PostC_2_RH (sub-723, one run), propagating noise-model uncertainty gives an event-HRF posterior mean of the expected shape. Its 95% credible band contains 0 at nearly all lags.

2. Interpretation (hypotheses, not tested here)

- **Spatial mismatch:** a single scalp channel (Cz) is a coarse proxy for the local activity that drives each cortical parcel's hemodynamics.
- **Little trial-to-trial variability:** the ERP area summarizes one component at one site and varies little around its mean. That gives the trial-wise model almost no leverage.
- **Shared non-neural signals:** broadband continuous EEG is dominated by noise, artifacts and slow drifts. Shared systemic or motion components could explain near-zero-lag "coupling" better than neurovascular coupling.
- **Power:** the EEG analyses have N = 5–6 subjects and ~80–125 mnt-correct trials each, against very small effects.

3. Implications and recommendations

1. **Treat the event-locked mnt-correct HRF as the primary fNIRS result.** So far, EEG-informed and EEG-weighted HRFs are null results.
2. **Require out-of-sample evidence** (held-out runs, permutation or phase-randomized-EEG nulls) before claiming EEG–fNIRS coupling. Parametric F-tests alone are not enough here.
3. **If the EEG question is pursued further:**
   - centre trial-wise features per subject (`a_i − ā`);
   - use features with physiological rationale and more trial variability (pre-stimulus posterior alpha, frontal theta, single-trial P3 amplitude);
   - use multichannel or source-level EEG matched to parcels;
   - pool across subjects with a hierarchical Bayesian model;
   - add subjects.
4. **Finish the open fixes first** (§8): rerun `city` after the alignment fix on the shared `Y_all`, fix the beta–basis pairing in `vis_hrf.py`, and rebuild sub-670's `Y_all`. These are unlikely to reverse the conclusions, but they remove known confounds.